



TRAINING PROGRAMME

RHCSA Part 1: Red Hat System Administration I

Run and manage a Red Hat Enterprise Linux 10 server

42 HOURS

FOR IT staff, technicians and students starting Linux system administration

PREPARES FOR RHCSA (EX200, RHEL 10), first half of the objectives



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Overview

This programme is the first half of the RHCSA path on Red Hat Enterprise Linux 10. Participants install RHEL, work at the command line, manage files, users, groups and permissions, control processes and systemd services, secure remote access with OpenSSH, keep logs and time correct, configure networking with NetworkManager and install software with dnf and Flatpak. Every chapter ends with a lab on RHEL 10 virtual machines, and the programme closes with a final project in which each participant builds and documents a server against a checklist. RHCSA Part 2 completes the exam objectives.

At a glance

#	PHASE	FOCUS	HOURS
01	Command Line Foundations	RHEL 10, help, text files and managing files	6 h
02	Users, Groups & Permissions	Local accounts, sudo, file permissions and special permissions	6 h
03	Processes & Services	Process control, systemd units and boot targets	6 h
04	Remote Access & Logging	OpenSSH, archives and transfers, logs and time	6 h
05	Networking, Software & Filesystems	NetworkManager, dnf and Flatpak, mounting and disk usage	6 h
06	Final Project	A documented RHEL 10 server, graded against a checklist	6 h
	Buffer	Catch-up, revision, extra lab time	6 h
	Total		42 h

Key facts

AUDIENCE	IT staff, technicians and students starting Linux system administration
PREREQUISITES	None. Basic computer use; no prior Linux experience needed
DURATION	42 h: 36 h of chapters + 6 h buffer
FORMAT	Short lectures followed by guided labs on RHEL 10
LAB ENVIRONMENT	Runs locally: one laptop per participant (16 GB RAM, hardware virtualization enabled) with two RHEL 10 virtual machines built from the trainer's images and registered with each participant's no-cost Red Hat Developer Subscription for Individuals; no cloud account or cost
ASSESSMENT	A lab validated at the end of each phase and a final project graded against a checklist; certificate of completion for participants who complete both
OUTCOME	Participants can install a RHEL 10 server and manage its users, permissions, services, SSH access, logs, network settings and software

PHASE 01

6 H

Command Line Foundations

Participants install RHEL 10, register it, and learn to find help and edit files without leaving the terminal. They then manage files and directories, links and wildcards. Validated lab: a freshly installed RHEL 10 virtual machine with a directory tree built, linked and documented entirely from the command line.

#	CHAPTER	CONTENT	FORMAT	HOURS
1.1	Getting Started with RHEL 10	Red Hat ecosystem, installation, subscriptions, first login, the web console (Cockpit)	Lecture + Lab	1.5 h
1.2	Help & Text Files	man, info, /usr/share/doc, vim, input and output redirection	Lecture + Lab	1.5 h
1.3	Command Line & Files	Bash, filesystem hierarchy, managing files, hard and symbolic links, globbing	Lecture + Lab	3 h

PHASE 02

6 H

Users, Groups & Permissions

Access control is the first job of an administrator. Participants create local users and groups, set password ageing, grant administrative rights through sudo, and control access to files with standard and special permissions. Validated lab: a team directory where group members can collaborate, other users cannot read, and one user can run selected commands through sudo.

#	CHAPTER	CONTENT	FORMAT	HOURS
2.1	Users & Groups	Local accounts, groups, passwords and ageing, privileged access with sudo	Lecture + Lab	3 h
2.2	File Permissions	User, group and other permissions, chmod and chown, special permissions (SUID, SGID, sticky bit), umask	Lecture + Lab	3 h

PHASE 03

6 H

Processes & Services

Participants observe and control what runs on the system: processes, jobs and signals, priorities, then the systemd units that start services at boot. Validated lab: a runaway process found and stopped, a service enabled at boot, and the default target changed and checked after a reboot.

#	CHAPTER	CONTENT	FORMAT	HOURS
3.1	Processes	Process states, jobs, signals, kill, top, load average, nice	Lecture + Lab	3 h
3.2	Services & systemd	Units, systemctl, enabling at boot, default target	Lecture + Lab	3 h

PHASE 04

6 H

Remote Access & Logging

Servers are administered remotely and diagnosed from their logs. Participants set up key-based SSH and harden the SSH server, archive and copy files between machines, make the journal persistent and keep time synchronised. Validated lab: key-only SSH access between the two lab machines, a compressed archive copied with rsync, and an event traced in the persistent journal.

#	CHAPTER	CONTENT	FORMAT	HOURS
4.1	OpenSSH	Remote access, key-based authentication, sshd hardening	Lecture + Lab	1.5 h
4.2	Archiving & Transfer	tar, gzip and bzip2, scp, sftp, rsync	Lecture + Lab	1.5 h
4.3	Logs & Time	journald, rsyslog, persistent journal, chrony	Lecture + Lab	3 h

PHASE 05

6 H

Networking, Software & Filesystems

Participants give the server a static network configuration, install and update software from RPM repositories and Flatpak, then mount filesystems and find where disk space goes. Validated lab: a server with a static IPv4 and IPv6 configuration and a hostname that survive a reboot, a package installed from a configured repository, and a filesystem mounted and checked.

#	CHAPTER	CONTENT	FORMAT	HOURS
5.1	Networking	NetworkManager, nmcli, IPv4 and IPv6 addresses, hostnames, name resolution	Lecture + Lab	3 h
5.2	Software	dnf, RPM repositories, RPM packages, Flatpak repositories and applications	Lecture + Lab	1.5 h
5.3	Filesystems	Mounting, block devices, df and du, locating files	Lecture + Lab	1.5 h

PHASE 06

6 H

Final Project

Each participant builds a RHEL 10 server from a written brief, reusing the configurations from the phase labs, then documents it and has it graded against a checklist.

#	CHAPTER	CONTENT	FORMAT	HOURS
6.1	Brief & Build	Read the brief, then configure users and groups policy, permissions, services, SSH hardening and networking	Project	3 h
6.2	Logging, Backup & Documentation	Persistent logging, archived backups copied to the second machine, a short administration guide	Project	1.5 h
6.3	Checklist Review	Reboot test, grading against the checklist, questions and feedback	Project	1.5 h

FINAL PROJECT

A RHEL 10 Server Ready for Service

A RHEL 10 server configured from a written brief: users and groups policy, file permissions, services enabled at boot, hardened SSH access, static networking, persistent logs and archived backups, all still correct after a reboot.

STACK RHEL 10, Bash, systemd, OpenSSH, NetworkManager, dnf, journald, rsyslog, chrony, tar, rsync

DELIVERABLE Configured server, administration guide, graded checklist